॥श्री गणेशाय नमः॥

Practice session 1

https://docs.google.com/forms/d/e/1FAIpQLScyshgOpEoak--Z_yepLsJUhUTrW-wrAZi6CfbNc_QojYvXiw/viewscore?vc=0&c=0&w=1&flr=0&viewscore=AE0zAgBPraxpXSwyugkx4jTFnPjBpv1UcPyIegW57yfWzIdLiDf0cjTjLsqs4tmdZCROWBQ

Q. In Simple Linear Regression, which of the following represents the formula for the Least Squares estimate of the slope β_1?

- A) $\frac{X_i^2}{Y_i}$
- B) $\frac{\sum(X_i-\bar{X})(Y_i-\bar{Y})}{\sum(X_i-\bar{X})^2}$ ✓
- C) $\frac{\sum X_i}{\sum Y_i}$
- D) $\sum X_i Y_i$

Q. Which of the following statements are true about the Least Squares method in Simple Linear Regression?

- A) It minimizes the sum of squared residuals. ✓
- B) It always results in an intercept of zero.
- C) It provides an unbiased estimate of the parameters β_0 and β_1. ✓
- D) It is sensitive to outliers. ✓

Q. Consider a Multiple Linear Regression (MLR) model with the following data for two independent variables X_1 and X_2, and the dependent variable Y:

X_1	X_2	Y
1	2	3
2	3	5
3	4	7
4	5	9

The regression equation is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

X is the design matrix, Y is the target variable, and $\beta = [\beta_0, \beta_1, \beta_2]^T$ is the coefficient vector.
Where λ is the regularization parameter and I is the identity matrix. Given this setup, which of the following statements are **true**?

- A) As λ increases, the estimated coefficients β_1 and β_2 will shrink. ✓
- B) When $\lambda = 0$, Ridge Regression is equivalent to Ordinary Least Squares (OLS). ✓
- C) Ridge Regression can lead to some coefficients being exactly zero, performing variable selection like Lasso.
- D) The Ridge solution is more stable when the predictors X_1 and X_2 are highly correlated (multicollinear). ✓

Q. In a Simple Linear Regression model, the following data is given:

$$X = [1, 2, 3, 4], \quad Y = [2, 4, 6, 8]$$

$y = 2x$

Using the Least Squares method, find the estimated slope β_1. (Round your answer to 1 decimal place.) 2.0

Q. In Gradient Descent for Simple Linear Regression, which parameter controls the step size for each iteration?

- A) Gradient
- B) Cost Function
- C) Learning Rate ✓
- D) Mean Squared Error

Q. Which of the following will affect the convergence of Gradient Descent in SLR?

- A) Learning Rate ✓
- B) Initial Values of Parameters ✓
- C) Number of Features ✓
- D) Size of the Dataset ✓

Q. In a regression model, the predicted values $\hat{Y} = [2, 3, 4, 5]$ and the actual values $Y = [1, 2, 5, 6]$. What is the Mean Squared Error (MSE)? (Round to 2 decimal places)

$MSE = \frac{1}{4}[(2-1)^2+(3-2)^2+(4-5)^2+(5-6)^2] = 1$

1.00

Q. Which of the following statements are correct regarding the coefficient of determination R^2?

- A) R^2 ranges between 0 and 1. ← R^2 can get -ve
- B) Higher R^2 values indicate a better fit of the model. ✓
- C) R^2 increases with the number of predictors, regardless of whether they are significant. ✓
- D) R^2 is unaffected by outliers.

Q. Which of the following are common assumptions made when performing statistical inference on Simple Linear Regression parameters?

- A) The errors are normally distributed. ✓
- B) The errors have constant variance (homoscedasticity). ✓
- C) The predictors are perfectly correlated.
- D) The relationship between predictor and outcome is linear. ✓

Q. In Multiple Linear Regression, the equation $y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$ assumes that:

- A) X_1 and X_2 are uncorrelated. ← a low correlation is fine as well
- B) X_1 and X_2 are perfectly correlated.
- C) The relationship between y and X_1, X_2 is linear. ✓
- D) $\beta_0 = 0$.

Q. Which of the following transformations are used in Polynomial Regression to capture non-linear relationships?

- A) Logarithmic transformation of X.
- B) Adding higher powers of X (e.g., X^2, X^3). ✓
- C) Taking the square root of X.
- D) Dividing X by the intercept.

Q. Which of the following are true about Ridge and Lasso regression?

- A) Ridge regression shrinks coefficients but does not perform feature selection. ✓
- B) Lasso regression can shrink coefficients to exactly zero. ✓
- C) Ridge regression minimizes $\sum |\beta_j|$.
- D) Both Ridge and Lasso are sensitive to the choice of the regularization parameter λ. ✓

Q. Consider a Simple Linear Regression (SLR) model with the equation:

$$\hat{y} = m \cdot x + c$$

You are using Gradient Descent to optimize the parameters m and c. The current values of the parameters are $m = 2$ and $c = 1$. Given that the learning rate $\eta = 0.1$, and the current gradients are $\frac{\partial J}{\partial m} = -4.5$ and $\frac{\partial J}{\partial c} = -2.0$, what will be the value of $(m + c)$ after one update using Gradient Descent?

(Provide your answer rounded to 2 decimal places.)

$m = m - \eta \frac{\partial J}{\partial m} = 2 - (0.1)(-4.5) = 2.45$

$c = c - \eta \frac{\partial J}{\partial c} = 1 - (0.1)(-2) = 1.2$

$2.45 + 1.2 = 3.65$

Q. Consider a Multiple Linear Regression (MLR) model with the following data for two independent variables X_1 and X_2, and the dependent variable Y:

X_1	X_2	Y
1	2	3
2	3	5
3	4	7
4	5	9

The regression equation is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

X is the design matrix, Y is the target variable, and $\beta = [\beta_0, \beta_1, \beta_2]^T$ is the coefficient vector.
Where λ is the regularization parameter and I is the identity matrix. Given this setup, which of the following statements are **true**?

- A) As λ increases, the estimated coefficients β_1 and β_2 will shrink. ✓
- B) When $\lambda = 0$, Ridge Regression is equivalent to Ordinary Least Squares (OLS). ✓
- C) Ridge Regression can lead to some coefficients being exactly zero, performing variable selection like Lasso.
- D) The Ridge solution is more stable when the predictors X_1 and X_2 are highly correlated (multicollinear). ✓

8.

Consider a Multiple Linear Regression (MLR) model with the following data for two independent variables X_1 and X_2, and the dependent variable Y:

X_1	X_2	Y
1	2	3
2	3	5
3	4	7
4	5	9

The regression equation is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

X is the design matrix, Y is the target variable, and $\beta = [\beta_0, \beta_1, \beta_2]^T$ is the coefficient vector. Given this setup, which of the following statements are **true**?

- A) The matrix $X^T X$ is singular, and thus the OLS solution does not exist. ✓
- B) The coefficients can still be estimated using Ridge Regression even if the matrix $X^T X$ is not invertible. ✓
- C) The values of β_0, β_1, and β_2 are $0, 1, 1$, respectively.
- D) Adding an additional feature to the model will resolve the issue of the matrix being singular.

$$X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 14 \\ 10 & 30 & 40 \\ 14 & 40 & 54 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1 - R_2} \begin{bmatrix} 4 & 10 & 14 \\ 10 & 30 & 40 \\ 0 & 0 & 0 \end{bmatrix}$$

$\det(X^T X) = 0$

$(X^T X - \lambda I)$ can be invertible

can't calculate values

$\det(X^T X)$ will still be zero